

# ECE 622: Modeling and Verification of Embedded Systems (Spring 2026)

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## 1. Reachability Analysis

The given UPPAAL model consists of two processes, **sys1** and **sys2**, that communicate using channel **c**.

The system also contains two clock variables, **x** and **y**.

- Process **sys1** has two locations: **A** and **B**
- Process **sys2** has two locations: **F** and **G**

We check reachability of two target conditions using UPPAAL queries.

### Condition 1

The first condition requires:

**sys1** to be in location **B**

**sys2** to be in location **F**

**x** = 1

**y** = 1

The UPPAAL query used is:

`E<> (sys1_s26.B and sys2_s26.F and sys1_s26.x == 1 and sys1_s26.y == 1)`

The model checker reports that this state **is reachable**.

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## Condition 2

The second condition requires:

- **sys1** to be in location **B**
- **sys2** to be in location **G**
- $x = 1$
- $y = 1$

The UPPAAL query used is:

$E \langle \rangle (\text{sys1\_s26.B and sys2\_s26.G and sys1\_s26.x} == 1 \text{ and sys1\_s26.y} == 1)$

The model checker reports that this state **is not reachable**.

Therefore,

- **Condition 1:** Reachable
- **Condition 2:** Not reachable

## 2. Shortest Path to the Reachable Condition

Since only **Condition 1** is reachable, we determine the **shortest path to reach this state**.

The shortest path requires **three jumps**. Initially the system starts in the following state:

- **sys1** at location **A**
- **sys2** at location **F**
- $x = 0$
- $y = 0$

Jump 1: During the first jump, the following synchronized transition occurs:

**sys2** moves from **F** → **G**

**sys1** takes its self-loop **A** → **A**

After the jump:

- **sys1** remains in location **A**
- **sys2** moves to location **G**
- **y** is reset to 0  $(2, 2) \rightarrow (2, 0)$
- Now we are on the line  $y = x - 2$

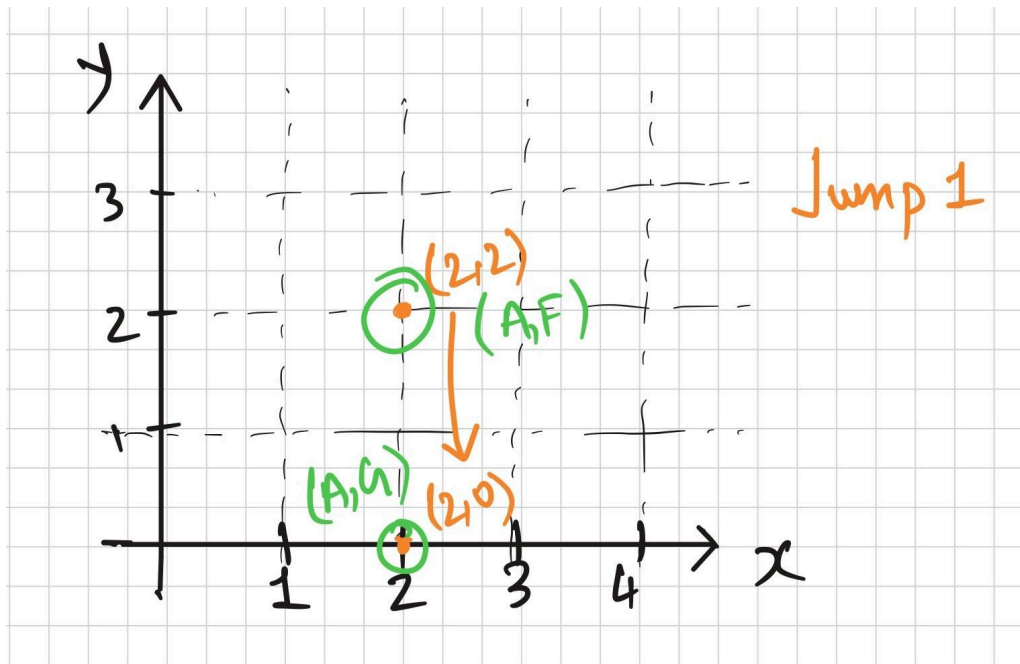


Figure 1. Clock region plot for Jump 1.

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Jump 2: In the second jump:

- **sys1** moves from **A**  $\rightarrow$  **B**
- **sys2** does **not change location** and remains in **G**

After the jump:

- **sys1** is now in **B**
- **sys2** remains in **G**
- Still on  $(2, 0)$

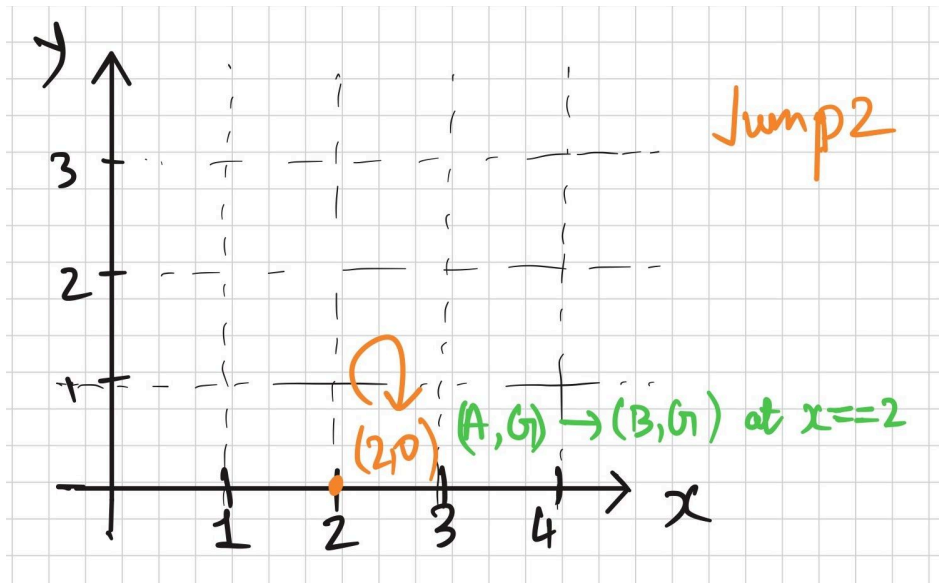


Figure 2. Clock region plot for Jump 2 to be inserted here.

Jump 3: In the third jump:

**sys2** moves from **G** → **F**

**sys1** takes its self-loop **B** → **B**

After the jump:

- $\text{sys1}$  in  $B$
- $\text{sys2}$  in  $F$
- $x = 1$
- $y = 1$

This satisfies **Condition 1**.

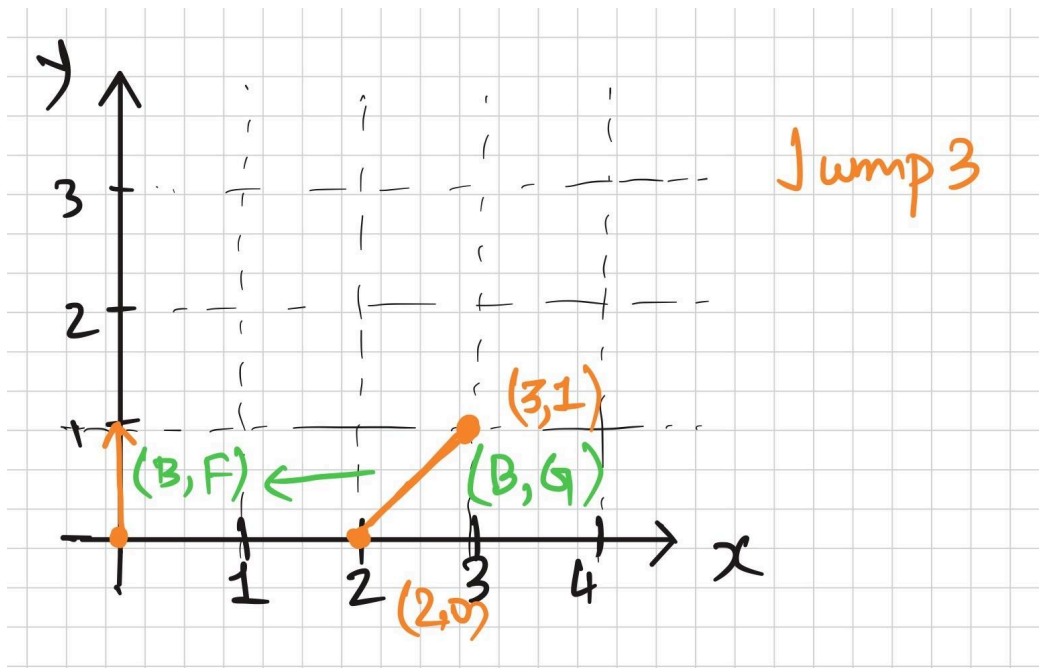


Figure 3. Clock region plot for Jump 3 to be inserted here.

### 3. Final Result

The reachability analysis gives the following results:

- **Condition 1:** Reachable
- **Condition 2:** Not reachable

The shortest execution path that satisfies Condition 1 consists of **three jumps**:

1. **sys2:**  $F \rightarrow G$  while **sys1** executes  $A \rightarrow A$
2. **sys1:**  $A \rightarrow B$  while **sys2** remains in G
3. **sys2:**  $G \rightarrow F$  while **sys1** executes  $B \rightarrow B$

After these transitions, the system reaches the target state with:

- $\text{sys1} = B$
- $\text{sys2} = F$
- $x = 1$
- $y = 1$